

Derek Serrano

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Backend Software Engineer with experience building scalable, distributed systems and customer-facing platforms using TypeScript/Node.js and Java.

EXPERIENCE

Software Engineer – Intact Insurance June 2024 – present

- Built and shipped customer-facing SaaS features using TypeScript and Java across a distributed architecture of 20+ microservices serving 50K+ users
- Owned development and production support for 3–5 backend services, collaborating with cross-functional teams to deliver bi-weekly releases for commercial insurance platforms
- Improved API performance by 30% by implementing Redis caching and Kafka-based asynchronous messaging across Kubernetes-hosted microservices
- Reduced MTTR by 35% and maintained 99.9% uptime by improving observability, alerting, and production diagnostics across distributed services
- Led migration of 10+ services from Spring Framework 5 to 6, resolving breaking changes and preventing production regressions during rollout
- Designed scalable REST APIs and internal service contracts, documenting interfaces with OpenAPI/Swagger to improve onboarding and integration reliability
- Designed and optimized PostgreSQL and MongoDB data models, defining schemas and improving query performance

Software Engineer Intern – Intact Insurance May 2022 – Aug 2023

- Contributed to backend service development and bug fixes within a Java-based microservices system
- Gained hands-on experience with CI/CD pipelines, code reviews, and production release processes in distributed systems

Machine Learning Research Assistant – Healthcare Systems R&A May 2021 – August 2021

- Collaborated with researchers and engineers to develop vision-based image processing tools for lab testing workflows
- Implemented and evaluated machine learning models including SVMs and neural networks, contributing to data preprocessing, model validation, and performance evaluation in experimental pipelines

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, C/C++

Frontend: React, Angular

Backend: Node.js, Spring Boot, REST APIs, GraphQL, Microservices

Infrastructure & Cloud: Kubernetes, Docker, AWS, OpenShift, Kafka, Redis

Databases: PostgreSQL, MongoDB

Tools: Jenkins, Git, Elasticsearch, Dynatrace

EDUCATION

Bachelor of Computer Engineering / McMaster University • 2019-2024 • GPA: 3.8